MCFNet: Multiscale Cross-Domain Fusion Network for HSI and LiDAR Data Joint Classification

Qiya Song, Feng Mo, Kexin Ding, Lin Xiao, Renwei Dian, *Member, IEEE*, Xudong Kang, *Senior Member, IEEE*, and Shutao Li, *Fellow, IEEE*

***Abstract*— The hyperspectral image (HSI) encompasses abundant spatial and spectral details, while light detection and ranging (LiDAR) delivers precise elevation data. The amalgamation of HSI and LiDAR data significantly improves the precision of image classification. However, most methods focus solely on spatial features while neglecting frequency-domain information, limiting the ability of deep models to characterize land cover. Furthermore, how to establish a sufficient interaction between different modalities is also an important issue. In this article, we propose a novel multiscale cross-domain fusion network (MCFNet) for joint classification of HSI and LiDAR data. The main idea is that the wavelet transform can provide details at different resolutions simultaneously, supplementing spatial-domain information and enriching feature representation. In addition, the multimodal fusion module (MFM) guided by HSI and the cross-domain fusion module (CDFM) strategy are developed to integrate features from diverse modalities and domains, respectively. Specifically, frequency-domain features are extracted by the discrete wavelet transform (DWT), and spatial-domain features of the image are captured through a set of convolutional operations. Then, interactive fusion is performed by MFM and CDFM, and finally, the integrated features are categorized using a classification module. Extensive experiments on three widely used HSI and LiDAR datasets indicate that MCFNet outperforms the state of the arts (SOTA) methods. The code will be available at: https://github.com/MSFLabX/MCFNet**

***Index Terms*— Hyperspectral images (HSIs), image classification, light detection and ranging (LiDAR) data, wavelet transform.**

I. Introduction

HYPERSPECTRAL image (HSI), due to its rich spectral information content, finds widespread applications across multiple domains, including change detection [1], [2], [3] and land cover classification. HSI-based land cover classification typically encompasses both conventional unimodal methods and multisource image-based methods [4], [5], [6], [7]. Among them, Tu et al. [5] performed classification by using a network based on three branches of superpixel–pixel–subpixel, while Tu et al. [6] addressed the problem of adversarial sample vulnerability in the model by modeling the global textual relationship of pixels. Bruni et al. [7] developed a wavelet-adaptive band selection framework that intelligently identifies and selects optimal spectral bands by analyzing class-specific spectral signatures in HSI. Wang et al. [8] proposed S2Mamba model, which employs dual scanning mechanisms with a selective state space model to capture long-range dependencies at linear complexity, replacing traditional self-attention. However, spectral information are not enough for some objects with similar categories; then, multisource image fusion classification methods have been explored. The elevation information from light detection and ranging (LiDAR) can complement the HSI to improve the effectiveness of classification. With the full fusion of two modalities, we can more accurately categorize the ground objects in some scenarios include, agricultural monitoring, marine monitoring, and mineral detection [9], [10].

In recent years, numerous fusion classification methods for HSI and LiDAR data have emerged, the extraction of rich features from images and the design of robust fusion methods [11], [12]. These methods typically fuse the spatial-domain information of these two modalities through early fusion, middle fusion, or late fusion approaches. For example, in [12], a dual-tributary architecture is developed to capture the features of HSI and LiDAR images, combining heterogeneous features during the fusion stage by employing feature-level and decision-level fusion strategies. Moreover, techniques such as Transformer have been widely used in various fields, proving the effectiveness of feature representation [13], [14], [15], [16], [17]. For instance, Ding et al. [9] combined Transformer and convolutional neural network (CNN) to model intricate local and global spatial–spectral relationships for the joint HSI and LiDAR data classification. Yao et al. [15] proposed ExViT, which is capable of handling multimodal remote sensing image patches with position-shared ViT parallel branches, extending the separable convolution module. Deng et al. [16] captured spatial features from HSI and SAR by combining a two-branch GCN with a Transformer. To improve feature discrimination, Lu et al. [17] leveraged the spectral reflectance characteristics of images and critical attributes of attention mechanisms. However, the aforementioned methods are based solely on the spatial domain of images, with minimal investigation of the frequency-domain features of ground object

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Qiya Song, Feng Mo, and Lin Xiao are with the School of Information Science and Engineering, Hunan Normal University, Changsha, Hunan 410081, China (e-mail: Mofeng711@163.com).

Kexin Ding, Xudong Kang, and Shutao Li are with the School of Robotics, Hunan University, Changsha, Hunan 410082, China.

Renwei Dian is with the School of Robotics, Hunan University, Changsha, Hunan 410082, China, and also with Fujian Key Laboratory of Spatial Information Perception and Intelligent Processing, Yango University, Fuzhou, Fujian 350015, China.

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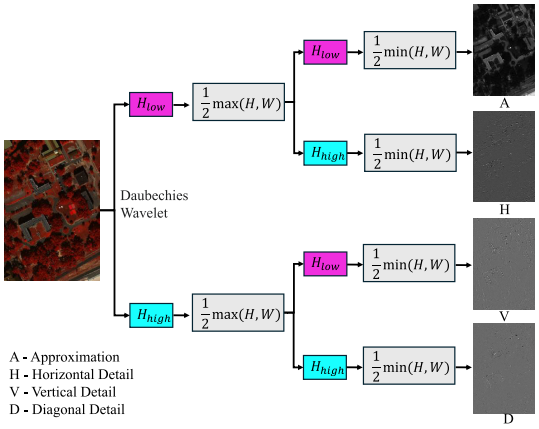


Fig. 1. Feature decomposition principle of Daubechies wavelet transform. H_{low} denotes the low-pass filtering of the original data, resulting in low-frequency approximation coefficients, while H_{high} refers to the high--pass filtering, producing high-frequency detail coefficients. The low- and high-frequency components of the image are extracted by convolutional operation, and four subbands are constructed to obtain one low frequency feature and three high frequency features. In this way, different frequency information of the image is decomposed and processed.

images. This constraint restricts the comprehensive utilization of deep models' representational potential in classification tasks.

With the progression of deep learning (DL), Fourier transform (FT) has gained increasingly significant in the realm of image processing, such as image super-resolution [18], [19], image dehazing [20], mirror detection [21], and hyperspectral anomaly detection [22], [23]. Zhao et al. [24] proposed a Transformer based on FT, which first performs pixel-level fusion of HSI and LiDAR images, then extracts global contextual and sequential features through the image Transformer, effectively achieving classification. Zeng et al. [25] introduced a method for adaptive fusion of cross-modal frequency-domain features, which separately fuses the amplitude and phase information of different modalities, alleviating the issue of modality differences. However, the methods that use FT for frequency-domain analysis have a limitation: FT can only focus on global frequency-domain analysis. In contrast, wavelet transform can simultaneously provide both frequency- and time-domain details, thereby complementing and enriching the local details of the image [26], [27]. Wavelet transform can extract features in a multiscale and multiresolution manner. As shown in Fig. 1, this is the process of the Daubechies wavelet transform, where H_{low} is the low-pass filter and H_{high} is the high-pass filter. In the first decomposition stage, the larger dimension between the length and width of the feature is halved, and low-pass and high-pass filter matrices are constructed to perform the 2-D wavelet transform, while in the second decomposition, the same operation is performed on the smaller dimension. The components of the image in low- and high-frequency bands are extracted by convolutional operations. Wavelet decomposition yields four distinct components: The components A, H, D, and V correspond to the low frequency, horizontal high-frequency, diagonal high-frequency, and vertical high-frequency elements, respectively. This method effectively captures structural details. Therefore, recent studies have attempted to use wavelet transform for the joint HSI and LiDAR classification. For example, Ni et al. [28] proposed a method that combines CNN, Transformer, and discrete wavelet transform (DWT), effectively improving the classification performance. The method initially employs 2-D and 3-D DWTs on HSI data and 2-D DWT on LiDAR data to extract both high- and low-frequency features from each modality. Then, it fuses the multimodal features and finally designs a frequency-domain-based global feature encoder for classification. Nevertheless, the above method still lack exploration of the spatial domain, which suppresses the capacity of the method to express spatial features.

By addressing the aforementioned challenges, this article proposes a novel multiscale cross-domain fusion network (MCFNet). The primary motivation for proposing MCFNet is to harness the robust spatial feature extraction (SFE) capabilities of CNNs and the strengths of wavelet transforms in capturing temporal and frequency-domain texture details, to achieve the superior global and local feature extraction. On the one hand, CNNs are employed to capture spatial-domain features from the presegmented multiscale inputs. On the other hand, the inputs are divided into one low-frequency group and three high-frequency groups, thereby capturing the texture and edge information of the image through wavelet transform. Subsequently, the multimodal fusion module (MFM) and the cross-domain fusion module (CDFM) are introduced to integrate features across different scales and domains. Our method demonstrates that the fusion of spatial and wavelet domains can significantly enhance classification performance.

The main contributions of our work are summarized as follows.

1) A framework for joint classification of HSI and LiDAR data is proposed by fusing spatial and wavelet domain features. This method effectively mines spatial- and frequency-domain information from images of different modalities and improves the overall classification performance.
2) To effectively capture multiscale features and complementary information across different modalities, we design CDFM and MFM. The CDFM is used to fuse spatial domain with wavelet domain information, while MFM exploits the spectral features of HSI data and the spatial structural details of LiDAR data.
3) Comprehensive experimental results validate the effectiveness of the proposed method. Furthermore, when evaluating the proposed method, the average results of ten trials in all comparative experiments outperform other state of the arts (SOTA) methods.

We structure the remainder of this article as follows. Section II reviews relevant literature on wavelet transform and DL-based fusion methods. Section III provides a detailed description of the proposed MCFNet. Section IV compares the MCFNet with SOTA methods and conducts comprehensive ablation studies to validate its effectiveness and evaluate its performance. Finally, Section V concludes this article with a summary of the key contributions and findings.

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II. Related Works

This section provides a detailed review of significant progress in image classification, broadly divided into frequency-domain-based approaches and multimodal fusion methods.

A. Frequency-Domain-Based Methods

The methods of frequency domain initially decompose the input image into four frequency components: one low-frequency group and three high-frequency groups. Subsequently, these low- and high-frequency components are fused for the following task. Zhao et al. [24] introduced supervised and adversarial losses designed in the Fourier space. By effectively combining complementary details from both frequency and spatial domains, which enhances the high-frequency detail recovery capability and perceptual quality of low-complexity generators in image super-resolution tasks. In [29], a wavelet transform-based frequency division interactive CNN (WFDI-CNN) is proposed to reduce spatial redundancy by decomposing the common tensor into low- and high-frequency complementary tensors. Li et al. [30] suppressed the aliasing effect and improve noise robustness by replacing traditional downsampling operations, such as maximum pooling and step-size convolution with the DWT operation.

Xiong et al. [31] developed a semi-supervised domain adaptation framework featuring a wavelet Siamese network that enables spatial–frequency interaction across bitemporal HSIs, significantly improving the change boundary detection accuracy. Xing et al. [32] proposed frequency-enhanced mamba for remote sensing change detection, developing a novel DCT-assisted Mamba decoder to achieve simultaneous feature decoding and refinement. Ni et al. [28] transformed the input data into different frequency intervals by DWT, while downsampling to obtain low- and high-frequency components, and learned deep features by 3-D DWT and 2-D DWT. Although these methods successfully harness frequency-domain information, they fail to establish a synergistic relationship between frequency components and their spatially correlated semantic patterns.

B. Multimodal Fusion Methods

With the advancement of DL, an increasing number of studies have focused on incorporating DL methods into multimodal tasks, facilitating remarkable progress in the domain of multimodal learning [33], [34]. The steps of multimodal fusion classification typically involve three key stages: feature extraction, fusion, and classification. HSI mainly emphasizes on the spectral resolution while LiDAR focuses more on the elevation information. Existing researches fuse these two types of features by designing various fusion methods. Hang et al. [12] selected the better modality features for fusion by using decision-level fusion, but the weights have a large impact, and inappropriate weights can interfere with the classification results. In [15], the classification tokens of all modalities are fused together by a multimodal attention and token fusion strategy before the head of the multilayer perceptron (MLP) to acquire the final classification distribution. Zhang et al. [35] introduced an oriented attention fusion (OAF) method, which extracts cross-modal shallow local shape features and integrates them along horizontal and vertical dimensions to enhance feature consistency. Lu et al. [10] proposed a multilevel fusion method, which combines different level features to generate the final classification results through an adaptive probabilistic fusion strategy. The above methods tend to overlook the hierarchical interdependence between these two modalities. In our experiment, we verify the main role of HSI in the fusion process and design a feature fusion method guided by HSI.

III. Proposed Method

In this article, we propose an MCFNet for HSI and LiDAR data, which integrates multiscale local spatial-domain details with multiscale global wavelet domain details to enhance classification accuracy. As illustrated in Fig. 2, the proposed MCFNet comprises three core parts: spatial-frequency feature extraction (SFFE), cross-domain fusion, and classification and loss function. This section elaborates on the design details of each component.

A. Spatial–Frequency Feature Extraction

Fig. 2 portrays the structure of SFFE. In general, the single-spatial-domain approach is deficient in detail texture feature extraction, while the tandem SFFE approach lacks the representation of complementary information. For this reason, we propose a multiscale, spatial–frequency-based feature extraction framework tailored for the two kinds of data. The SFFE module comprises two key components: SFE and wavelet feature extraction (WFE). Briefly, the SFE acquires the local spatial details of images, and the WFE acquires the detailed texture features through 2-D DWT. Then, we capitalize on local spatial details and global texture features at multiscale.

In the multimodal fusion classification task, there is a complex process of feature extraction and fusion. Given two different modalities, HSI and LiDAR, denote as $X_h \in \mathbb{R}^{m\times n\times d_h}$ and $X_l \in \mathbb{R}^{m\times n\times d_l}$, where m and n related to the height and width of HSI and LiDAR data, respectively. d_h and d_l denote the number of spectral bands of them, respectively. Our task is to learn reliable feature representations from these modalities for classification.

1) Wavelet Feature Extraction: The network composition of WFE can be simply observed in Fig. 2(a). WFE is divided into three main steps: wavelet transform, convolutional coding, and frequency-domain feature alignment (FDFA). Firstly, we segment the input image into two different-sized patches, 8×8 and 16×16. After padding the edges of the image, patches are extracted for each pixel to generate HSI cubes $X_h^p \in \mathbb{R}^{p\times p\times d_h}$ and LiDAR cube $X_l^p \in \mathbb{R}^{p\times p\times d_l}$, where p is the space size of the cubes. Then, the image is decomposed into three high frequency constituents (X_h^{HL}, X_h^{LH}, X_h^{HH}) and one low frequency constituent (X_h^{LL}) by wavelet transform, this operation allows the model to learn detailed features from the high-frequency components. Subsequently, we will get

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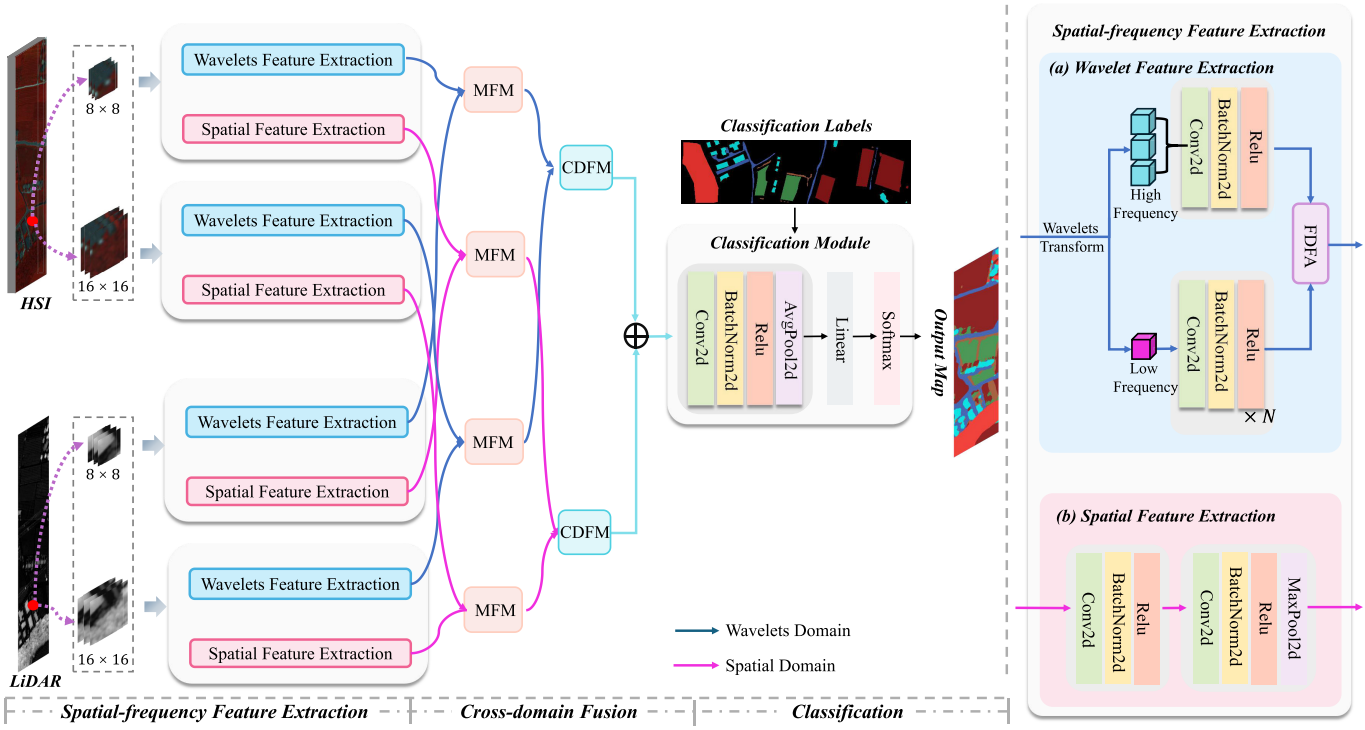


Fig. 2. Proposed framework of the MCFNet method for HSI and LiDAR data joint classification. First, the HSI and LiDAR images are divided into two different sizes for WFE and spatial-domain feature extraction, respectively. Also, feature fusion is carried out through the MFM guided by the features of HSI. Subsequently, it is input to the CDFM to fuse the spatial-domain and frequency-domain features, and finally to the classification module for joint multimodal classification. (a) Frequency-domain features are extracted using DWT. (b) Spatial and features are extracted with several sets of convolutional operations.

the components separately for convolutional coding. Then, we apply convolutional encoding to the components separately and fuse the high-frequency and low-frequency features through FDFA. Specifically, for the low-frequency components containing structural features, we directly apply multilayer deep convolutions for extraction

$$X_h^1 = \xi\left(\mathrm{BN}\left(\mathrm{Conv}\left(X_h^{LL}\right)\right)\right) \tag{1}$$

$$X_h^2 = \xi\left(\mathrm{BN}\left(\mathrm{Conv}\left(X_h^1\right)\right)\right) \tag{2}$$

$$X_h^3 = \mathrm{Concat}\left(\mathrm{mean}\left(X_h^2 0\right), \max\left(X_h^2\right)\right) \tag{3}$$

$$X_h^l = \sigma\left(\mathrm{BN}\left(\mathrm{Conv2}d\left(X_h^3\right)\right)\right) \times X_h^3 \tag{4}$$

where σ means sigmoid function, $\xi(\cdot)$ denotes the ReLU function, and BN represents batch normalization operation. Correspondingly, we obtain the low-frequency features X_l^l of the LiDAR image after similar processing. We first combine the three high-frequency constituents and then perform convolution on the input feature maps, using batch normalization to stabilize the learning process and applying the ReLU activation function to introduce nonlinearity, in order to capture detailed features

$$X_h^h = \xi\left(\mathrm{BN}\left(\mathrm{Conv}\left(\mathrm{Cat}\left(X_h^{LH}, X_h^{HL}, X_h^{HH}\right)\right)\right)\right). \tag{5}$$

Equivalently, we obtain the high-frequency features X_l^h of the LiDAR image after similar processing. Subsequently, we design a FDFA module to align and fuse the high- and low-frequency features. High-frequency and low-frequency features capture distinct aspects and attributes of the image. However, due to their inherent semantic differences, semantic

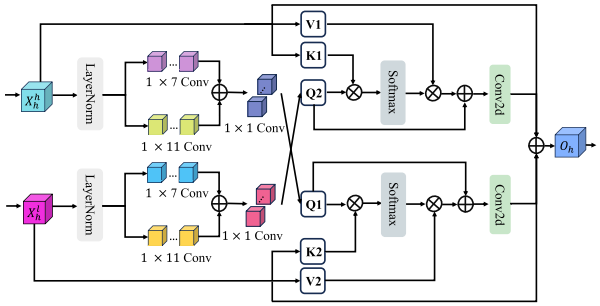


Fig. 3. Illustration of our FDFA. This structure is used for semantic alignment and feature fusion between high-frequency and low-frequency components. We first perform mappings with kernel sizes of 1×7 and 1×11 on these two types of features to adjust them to a unified scale. Then, cross attention is utilized to enable interaction and selection between the two features. Finally, the selected features are combined.

alignment is necessary to ensure that the image representation remains complementary. Inspired by [36], we design the FDFA. As shown in Fig. 3, first, the previously obtained X_h^l and X_h^h are passed through a layer normalization operation, and then, the resulting features undergo multiscale convolution and weighted summation to obtain the q value. Given the symmetric architecture, we take X_h^h as an illustrative example, where both V and K are obtained directly from the representation of X_h^h. The process involves computing the matrix multiplication between the swapped Q_2 and K, applying Softmax activation to the resulting attention weights, then multiplying with V. This output is combined with the original Q_2 through residual addition, and finally processed by

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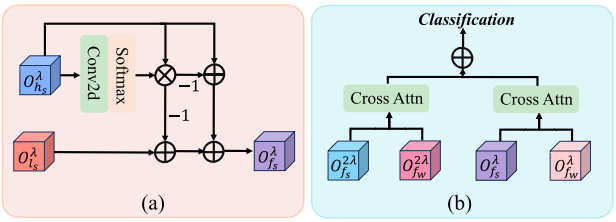


Fig. 4. Illustration of (a) MFM and (b) CDFM.

a 2-D convolutional layer to perform weighted fusion with the low-frequency features. Through this way, the aligned features O_h is obtained with enhanced representational consistency. The same processing approach applies to other scales and LiDAR data.

2) Spatial Feature Extraction: By extracting CNN-based spatial-domain features, we effectively capture localized image details. The encoding architecture comprises two sequential modules, the first one incorporating a 3 × 3 2-D convolutional layer, which is configured with 32 fixed kernels. This hierarchical feature extraction process can be mathematically formulated as follows:

$$X_h^s = \xi\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(X_h^p\right)\right)\right) \tag{6}$$

$$X_l^s = \xi\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(X_l^p\right)\right)\right) \tag{7}$$

where X_h^s and X_l^s denote the spatial features of HSI and LiDAR. To halve the spatial dimensions of the feature cube, a max-pooling operation is strategically incorporated in the second encoding block. This design serves dual purposes: 1) progressively expanding the receptive field through hierarchical downsampling and 2) counteracting potential feature invariance artifacts induced by successive convolutional layers. The kernel quantity in the second layer is designated as a tunable hyperparameter d_f, whose empirical determination will be elaborated in Section IV. The proposed methodology is expressed by the following mathematical representation:

$$\hat{X}_h^s = \mathrm{Max\ Pool}\left(\xi\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(X_h^s\right)\right)\right)\right) \tag{8}$$

$$\hat{X}_l^s = \mathrm{Max\ Pool}\left(\xi\left(\mathrm{BN}\left(\mathrm{Conv}_{3\times3}\left(X_l^s\right)\right)\right)\right) \tag{9}$$

where $\hat{X}_h^s \in \mathbb{R}^{p\times p\times d_f}$ and $\hat{X}_l^s \in \mathbb{R}^{p\times p\times d_f}$ denote the HSI and LiDAR features extracted through spatial-domain feature extraction, respectively.

The multimodal feature extraction pipeline now yields cross-domain representations: 1) spatially encoded features preserving local geometric structures and 2) wavelet-transformed features capturing multiresolution spectral signatures.

B. Cross-Domain Fusion

To accurately capture image features across different modalities, we design a fusion mechanism guided by HSI named MFM. This mechanism allows for the flexible allocation of fusion weights based on HSI features, aiming to preserve maximal mutual data between HSI features and the fused multimodal features. As shown in Fig. 4(a), taking the spatial-domain features $O_{h_s}^{\lambda}$ and $O_{l_s}^{\lambda}$ with a patch size of 8 ($\lambda = 8$) as an example, we illustrate the details using the following formula:

$$\begin{aligned} O_{f_s}^{\lambda} &= O_{h_s}^{\lambda} - \left(\varsigma\left(\mathrm{Conv}\left(O_{h_s}^{\lambda}\right)\right) \times O_{h_s}^{\lambda}\right) \\ &\quad + O_{l_s}^{\lambda} - \left(\varsigma\left(\mathrm{Conv}\left(O_{h_s}^{\lambda}\right)\right) \times O_{h_s}^{\lambda}\right) \\ &= O_{h_s}^{\lambda} + O_{l_s}^{\lambda} - 2\left(\varsigma\left(\mathrm{Conv}\left(O_{h_s}^{\lambda}\right)\right) \times O_{h_s}^{\lambda}\right) \end{aligned} \tag{10}$$

where $O_{f_s}^{\lambda}$ denotes the output of MFM. Correspondingly, we obtain the fused feature $O_{f_w}^{\lambda}$ in the wavelet domain with a patch size of 8, the fused feature $O_{f_s}^{2\lambda}$ in the spatial domain with a patch size of 16, and the fused feature $O_{f_w}^{2\lambda}$ in the wavelet domain with a patch size of 16. Then, we fuse the features from the wavelet domain and the spatial domain through CDFM

$$\begin{aligned} \mathrm{Out} &= W_1\left(f\left(\text{Cross Attn}\left(O_{f_s}^{\lambda}, O_{f_w}^{\lambda}\right)\right)\right) + b_1 \\ &\quad + W_2\left(f\left(\text{Cross Attn}\left(O_{f_s}^{2\lambda}, O_{f_w}^{2\lambda}\right)\right)\right) + b_2 \end{aligned} \tag{11}$$

where $W_1 \in \mathbb{R}^{d_f\times p^2}$, $W_2 \in \mathbb{R}^{d_f\times p^2}$, $b_1 \in \mathbb{R}^{d_f}$, and $b_2 \in \mathbb{R}^{d_f}$ are learnable parameters. $f(.)$ means the flatten operation and d_f is the output dimension, which will be explained in detail in Section IV-E3.

C. Classification and Loss Function

The final classification module operates in two stages. First, a 2-D convolution with a 1 × 1 kernel is applied, followed by batch normalization, an ReLU activation, and adaptive average pooling. Second, The spectral feature vector undergoes a linear transformation $W^T x + b$ via learnable weights $W \in R^{(C\times C)}$ and bias $b \in R^C$, followed by Softmax normalization to produce class C membership probabilities (P), enabling probabilistic interpretation of pixelwise classification decisions. The classification loss $\mathcal{L}_{\mathrm{cls}}$ is computed using cross entropy, as defined by the following formula:

$$\mathcal{L}_{\mathrm{cls}}(P, Q) = -\sum_{i=1}^{C} P(x_i) \log Q(x_i) \tag{12}$$

where $\mathcal{L}_{\mathrm{cls}}(P, Q)$ represents the final classification probability, with $P(x_i)$ being the predicted probability of a pixel belonging to class i, and $Q(x_i)$ representing the expected probability of the pixel being classified as class i.

IV. Experiments and Analysis

In this section, we present a comprehensive overview of the datasets, metrics, and other experimental settings utilized in the work. In addition, we perform comparative and ablation experiments to illustrate the effectiveness and advancements of the MCFNet.

A. Dataset Description

1) Houston2013 Dataset: As a standard benchmark in urban remote sensing, the Houston2013 dataset provides paired HSI–LiDAR data with precise geospatial alignment. The hyperspectral cube captures 144 spectral channels, while the LiDAR-derived DSM offers complementary vertical structural information, both with 349 × 1905 pixels dimensions [37]. With 15 029 validated reference samples spanning 15 urban land cover classes. Table I details the training and testing sample distribution.

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TABLE I
QUANTITATIVE RESULTS ON HOUSTON2013 DATASET (OA, AA, KAPPA, AND CLASSWISE ACCURACY). BEST RESULTS ARE IN BOLD

No.	Class(Train/Test)	CoupledCNN	GLT	Fusion_HCT	AMSSE	ExViT	HFSNet	HyperMLP	FDNet	M2FNet	MCFNet
1	Healthy Grass(20/1231)	94.03	92.75	92.99	**96.51**	92.85	93.74	91.45	95.29	92.45	93.83
2	Stressed Grass(20/1234)	**97.46**	97.07	96.06	87.20	96.26	95.05	93.65	97.38	96.82	96.55
3	Synthetis Grass(20/677)	99.72	99.25	99.53	99.85	98.92	**99.93**	99.73	99.79	99.62	99.69
4	Tree(20/1224)	98.59	97.26	96.91	**100**	96.42	97.28	98.65	97.27	95.38	97.64
5	Soil(20/1222)	99.78	99.65	99.96	99.92	99.00	99.55	97.54	99.78	97.60	**99.99**
6	Water(20/305)	**98.79**	96.46	97.93	85.57	96.36	93.84	94.66	96.07	96.95	96.85
7	Residential(20/1248)	95.07	**95.36**	89.39	92.63	93.62	89.21	90.99	93.44	94.01	95.09
8	Commercial(20/1224)	89.04	**90.62**	85.10	84.64	88.72	84.20	86.24	82.78	90.07	89.22
9	Road(20/1232)	85.81	91.78	81.40	87.99	91.40	79.18	84.52	80.67	91.48	**93.99**
10	Highway(20/1207)	93.02	91.43	**96.66**	77.22	91.11	83.14	86.31	92.35	93.17	93.12
11	Railway(20/1215)	95.41	94.19	**95.90**	93.74	93.83	92.00	77.81	90.39	93.60	95.14
12	Park Lot 1(20/1213)	87.32	90.50	85.02	81.53	87.42	87.56	76.29	87.87	86.43	**92.08**
13	Park Lot 2(20/449)	98.06	97.66	98.13	86.41	97.22	86.10	86.30	92.36	95.95	**98.15**
14	Tennis Court(20/408)	**100**	99.93	99.98	99.26	99.63	99.78	98.16	99.90	99.93	**100**
15	Running Track(20/640)	99.45	99.98	**100**	99.69	99.69	99.86	99.91	99.72	99.81	**100**
	OA(%)	94.52	94.88	93.16	91.07	94.00	91.20	89.74	92.78	94.05	**95.43**
	AA(%)	95.44	95.59	94.33	91.48	94.82	92.03	90.82	93.67	94.88	**96.09**
	Kappa(%)	94.00	94.47	92.61	90.34	93.52	90.48	88.91	92.19	93.57	**95.06**

2) Muufl Dataset: The Muufl Gulfport dataset (2010, MS, USA) combines HSI from the ITRES CASI-1500 sensor (64 bands and 380–1050 nm) and LiDAR from an ALTM system (1064 nm). The HSI has anisotropic resolution (0.54 m cross-track × 1.0 m along-track), while LiDAR products have 0.60 × 0.78 m resolution. Both share spatial dimensions (325 × 220 pixels) and include 53 687 georeferenced samples across 11 land cover classes Table II, with classwise statistics in [39].

3) Trento Dataset: The 2015 Trento dataset offers co-registered HSI and LiDAR data for precision agriculture. The HSI cube (AISA Eagle sensor) includes 63 bands (420–990 nm), and the LiDAR-derived DSM (Optech ALTM 3100EA) provides structural data at 1-m resolution. With 30 214 validated samples across six classes, it is a key benchmark for agricultural remote sensing [38]. Sample distributions are detailed in Table III.

B. Experimental Setting

Our experiments are conducted on an Intel Xeon Platinum 8358 machine with an NVIDIA A800 GPU. The setup is implemented in PyTorch, using the Adam optimizer with a learning rate decay of 0.9. Based on empirical validation, we set a batch size of 64, a learning rate of 0.0005, and 500 training epochs, with evaluation every 20 epochs. Preprocessing involves generating fixed-size patches of 8 × 8 and 16 × 16 pixels. Model performance is assessed using four metrics: average accuracy (AA), per-category accuracy (CA), Kappa coefficient, and overall accuracy (OA). All metrics are averaged over ten independent trials for robustness and fairness.

C. Performance Comparison

In this section, we divide the description into comparison of different classification methods in three datasets.

1) Comparison of Different Classification Methods: To comprehensively evaluate the proposed MCFNet, we select nine SOTA multimodal remote sensing classification methods for comparison. These include CoupledCNN [12], GLTNet [9], FusionHCT [40], AMSSE [41], ExViT [15], HFSNet [42], HyperMLP [43], FDNet [28], and M2FNet [44].

Specifically, Hang et al. [12] proposed an efficient coupled CNN framework that synergistically integrates hyperspectral and LiDAR data through feature-level and decision-level fusion. Ding et al. [9] proposed a GLT-Net to integrate local SFE with global spectral dependency modeling. Zhao et al. [40] developed a dual-branch architecture that combines CNN-based feature extraction with Transformer-enabled feature learning, achieving comprehensive feature representation mining. Gao et al. [41] proposed an adaptive spatial–spectral network that enhances spectral correlation while preserving fine details through multiscale feature extraction. Yao et al. [15] integrated multimodal remote sensing data through separable convolution modules, cross-modal attention mechanisms, and classification token fusion strategies to harness complementary information. Song et al. [42] proposed an elevation-guided hierarchical fusion-and-separation network that employs deformable convolutions and edge decoders to mutually enhance modality-shared and modality-specific features. Li et al. [43] applied the MLP-Mixer to remote sensing image classification. Ni et al. [28] integrated DWT, fast FT, and deformable convolutional operations to combine local and global representations. Sun et al. [44] proposed a two-branch network through synergistic integration of multiscale 3-D–2-D mixed CNNs and lightweight Transformer to achieve feature mining.

Tables I–III provided a summary of sample configurations and classification performance across the three datasets. The tables employ a standardized format with optimal results in each category emphasized through bold formatting, facilitating direct comparison of method effectiveness under consistent evaluation protocols. Figs. 5–7 present a standardized visualization protocol across all datasets, sequentially displaying: (a) hyperspectral false-color images [bands 64 (R), 43 (G), 22 (B)], (b) LiDAR visualizations, (c) ground truth maps,

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TABLE II

QUANTITATIVE RESULTS ON MUUFL DATASET (OA, AA, KAPPA, AND CLASSWISE ACCURACY). BEST RESULTS IN BOLD

No.	Class(Train/Test)	CoupledCNN	GLTNet	FusionHCT	AMSSE	ExViT	HFSNet	HyperMLP	FDNet	M2FNet	MCFNet
1	Trees(20/23226)	85.68	90.33	85.77	78.69	88.05	87.44	84.99	85.68	**91.44**	89.96
2	Mostly Grass(20/4250)	75.70	74.51	76.49	60.26	73.23	74.14	**83.52**	75.70	70.55	76.05
3	Mixed Ground Surface(20/6862)	62.61	67.65	59.80	57.00	65.57	63.08	66.37	62.61	62.67	**74.10**
4	Dirt and Sand(20/1806)	82.10	83.78	86.69	84.22	86.22	82.85	84.96	82.10	84.25	**88.23**
5	Road(20/6667)	82.24	82.46	72.24	73.53	74.77	84.18	81.58	82.24	77.59	**87.17**
6	Water(20/446)	98.32	99.28	99.17	**100**	99.22	98.45	99.91	98.32	98.16	99.60
7	Building Shadow(20/2213)	88.41	89.17	84.42	89.02	86.47	88.81	82.87	88.41	83.67	**90.84**
8	Building(20/6220)	93.45	93.33	89.67	90.53	93.47	93.66	95.12	93.45	92.47	**95.23**
9	Sidewalk(20/1365)	**67.57**	64.47	52.72	54.07	53.25	66.57	47.22	67.57	55.41	46.56
10	Yellow Curb(20/163)	**82.76**	70.74	77.73	64.42	73.19	74.66	73.87	**82.76**	69.45	59.63
11	Cloth Panels(20/249)	96.06	95.18	95.54	**98.80**	94.46	92.25	96.63	96.06	94.46	95.78
	OA(%)	82.08	84.64	74.08	75.39	82.02	83.02	82.33	82.08	83.00	**85.97**
	AA(%)	**83.18**	82.81	80.02	77.32	80.71	82.37	81.55	83.18	80.01	82.11
	Kappa(%)	77.10	80.14	74.08	68.98	76.86	78.16	77.40	77.10	77.94	**81.80**

TABLE III

QUANTITATIVE RESULTS ON TRENTO DATASET (OA, AA, KAPPA, AND CLASSWISE ACCURACY). BEST RESULTS IN BOLD

No.	Class(Train/Test)	CoupledCNN	GLTNet	FusionHCT	AMSSE	ExViT	HFSNet	HyperMLP	FDNet	M2FNet	MCFNet
1	Apple Trees(20/4014)	98.94	99.18	98.87	**99.58**	98.94	97.42	94.51	99.07	99.39	99.29
2	Buildings(20/2883)	96.49	98.11	94.09	97.57	96.81	97.99	96.59	97.41	97.07	**98.45**
3	Ground(20/459)	97.86	98.58	98.17	96.51	98.61	97.02	95.29	**99.48**	97.97	98.47
4	Woods(20/9103)	99.90	**100**	**100**	**100**	**100**	99.91	99.61	99.88	99.95	**100**
5	Vineyard(20/10481)	99.96	**100**	98.78	99.95	99.82	99.22	98.58	98.94	99.96	**100**
6	Roads(20/3154)	97.12	96.63	93.75	**97.34**	96.68	95.90	95.90	96.39	96.36	96.79
	OA(%)	99.16	99.33	98.17	99.36	99.12	98.69	97.83	98.84	99.20	**99.40**
	AA(%)	98.38	98.75	97.27	98.49	98.47	97.91	96.75	98.53	98.45	**98.84**
	Kappa(%)	98.87	99.11	97.57	99.15	98.83	98.25	97.10	98.45	98.93	**99.20**

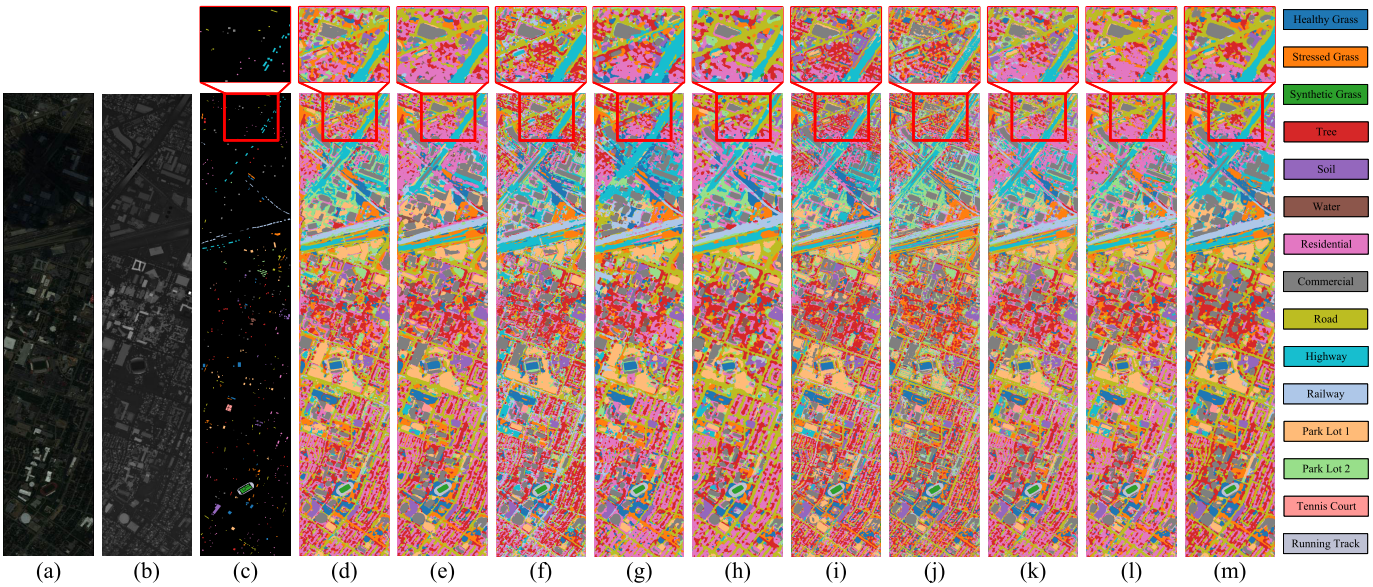


Fig. 5. Classification maps for all pixels in the Houston2013 dataset using different methods. (a) False-color HSI (using bands 64, 43, and 22, as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Ground-truth map. (d) CoupledCNN (94.52%). (e) GLTNet (94.88%). (f) FusionHCT (93.16%). (g) AMSSE (91.06%). (h) ExViT (94.00%). (i) HFSNet (91.20%). (j) HyperMLP (89.74%). (k) FDNet (92.78%). (l) M2FNet (94.05%). (m) MCFNet (95.43%).

and (d)–(m) comparative classification visualizations. This systematic presentation enables direct visual assessment of method performance relative to input data characteristics and ground reality. Then, a detailed analysis of these datasets follows.

2) Houston2013 Dataset: Our experiments, both quantitative and qualitative, demonstrate that MCFNet significantly outperforms competing methods across all evaluation metrics. As shown in Table I, MCFNet achieves average improvements of 0.55 in OA, 0.5 in AA, and 0.59 in Kappa coefficient

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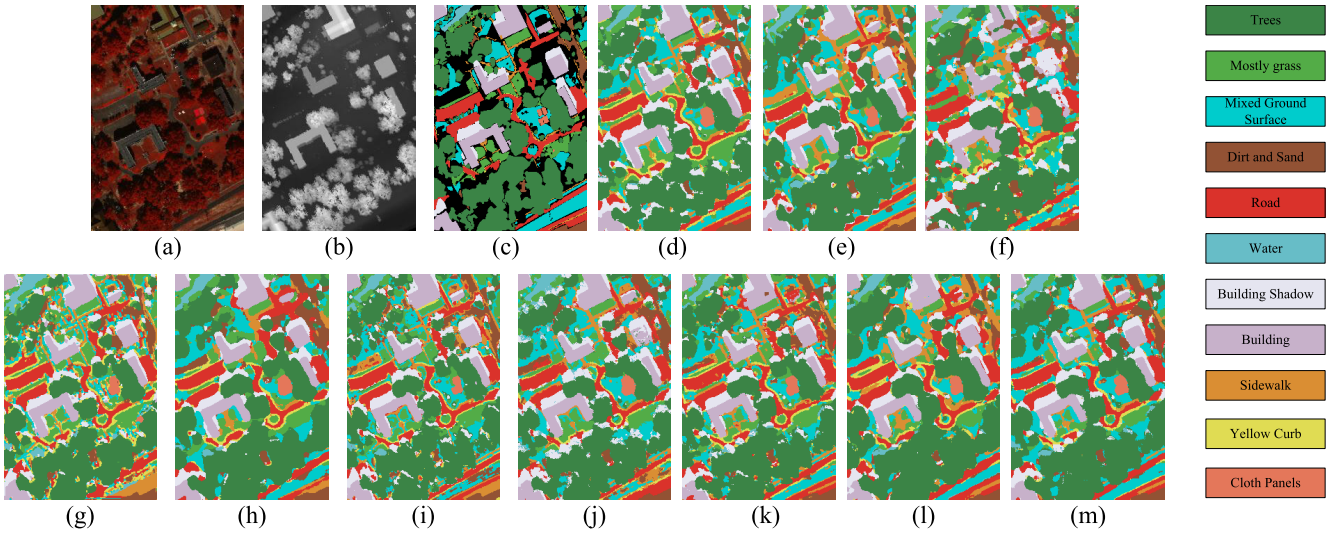


Fig. 6. Classification maps for all pixels in the Muufl dataset using different methods. (a) False-color HSI (using bands 40, 20, and 10, as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Ground-truth map. (d) CoupledCNN (82.08%). (e) GLTNet (84.64%). (f) FusionHCT (74.08%). (g) AMSSE (75.39%). (h) ExViT (82.02%). (i) HFSNet (83.02%). (j) HyperMLP (82.33%). (k) FDNet (82.08%). (l) M2FNet (83.00%). (m) MCFNet (85.97%).

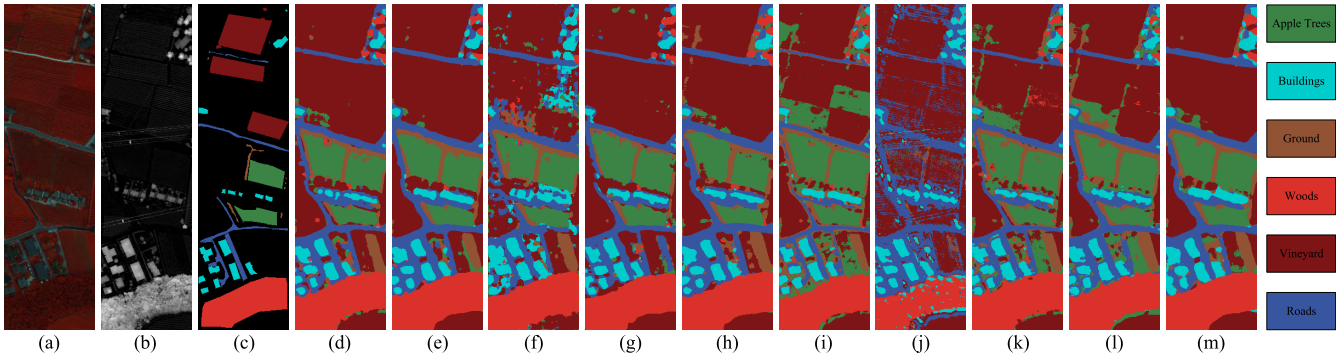


Fig. 7. Classification maps for all pixels in the Trento dataset using different methods. (a) False-color HSI (using bands 40, 20, and 10, as R, G, and B, respectively). (b) Gray image for LiDAR. (c) Ground-truth map. (d) CoupledCNN (99.16%). (e) GLTNet (99.33%). (f) FusionHCT (98.17%). (g) AMSSE (99.36%). (h) ExViT (99.12%). (i) HFSNet (98.69%). (j) HyperMLP (97.83%). (k) FDNet (98.84%). (l) M2FNet (99.20%). (m) MCFNet (99.40%).

compared to the suboptimal method, while attaining the highest classification accuracy for 6 out of 15 categories. Among other models, FDNet outperforms AMSSE and HFSNet, highlighting the effectiveness of parallel processing of frequency and spatial domains to capture both texture and spatial details. Fig. 5, generated by classifying all pixel points, shows that MCFNet accurately delineates edge contours and performs well in cloud- and fog-obscured areas, underscoring its robustness for classification tasks.

3) Muufl Dataset: Through quantitative experiments on the Muufl dataset, we find that the proposed model achieved statistically significant improvements, with an OA increase of 1.57%. Furthermore, it achieves superior classification performance in 5 of the 11 categories, surpassing other methods. For the high-spatial-resolution Muufl dataset, as shown in Table II, some methods exhibit results contrary to those on the Houston dataset. For instance, CoupledCNN ranks third on the Houston dataset but drops to fifth on the Muufl dataset, while models surpassing it, such as MCFNet, GLTNet, HFSNet, and HyperMLP, incorporated global awareness into the network without exception. Furthermore, in qualitative experiments, MCFNet also demonstrate outstanding performance. As shown in Fig. 6, MCFNet provides clearer boundary delineation for categories such as Building and Building Shadow, further validating the reliability of the results in Table II.

4) Trento Dataset: Table III summarizes the classification performance on the Trento dataset. MCFNet achieves the best performance across three metrics and attains the highest accuracy in three out of six categories. Fig. 7 illustrates the visualization of classification results on the Trento dataset in MCFNet. By leveraging the proposed MFM and CDFM to better integrate HSI and LiDAR features, our classification results for the Trento dataset appear smoother and less noise, aligning with the quantitative findings. Notably, while existing methods frequently misclassify Ground and Woods, MCFNet successfully distinguishes between them, proving its enhanced discriminative capability.

D. Ablation Studies

1) Effects of Different Module: To evaluate the impact of each module and ensure fairness, we replace each module with a corresponding convolutional module, maintaining a similar parameter scale. The modified models are denoted as "w/o CDFM," "w/o MFM," and "w/o SFFE." Table IV shows the

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TABLE IV
QUANTITATIVE COMPARISON OF THE SUBMODULES IN THE PROPOSED MCFNET

Method	Params(K)	Computation Cost(MFLOPs)	Houston2013			Muufl			Trento		
			OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)
w/o CDFM	461.60	63.61	94.35	95.03	93.89	84.46	81.49	79.92	99.36	98.72	99.14
w/o MFM	482.72	68.95	95.10	95.76	94.70	85.28	81.02	80.95	99.28	98.64	99.04
w/o SFFE	292.91	82.64	94.82	95.54	94.40	85.53	**84.24**	81.33	99.33	98.72	99.10
ours	549.02	39.06	**95.43**	**96.09**	**95.06**	**85.97**	82.11	**81.80**	**99.40**	**98.84**	**99.20**

TABLE V
QUANTITATIVE COMPARISON OF MODALITY MISSING IN MCFNET

Modality Methods	Params(K)	Computation Cost(MFLOPs)	Houston2013			Muufl			Trento		
			OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)	OA(%)	AA(%)	Kappa(%)
w/ HSI	401.14	48.14	93.34	94.18	92.80	83.46	82.94	78.63	98.15	96.70	97.53
w/ LiDAR	225.46	27.27	52.18	55.48	48.68	58.13	43.94	46.51	97.77	94.11	97.02
ours	549.02	39.06	**95.43**	**96.09**	**95.06**	**85.97**	82.11	**81.80**	**99.40**	**98.84**	**99.20**
LRDTN	148.05	7.81	87.32	88.96	86.29	82.04	77.34	76.69	97.72	96.20	96.97
MamTrans	2028.94	16.83	92.86	93.98	92.28	82.71	**84.73**	77.87	97.55	96.09	96.72

experimental results on three datasets using three metrics. The CDFM module has the most significant impact on the Houston2013 and Muufl datasets, highlighting its role in facilitating cross-domain feature interaction and enhancing information complementarity. In previous works, HSI and LiDAR features are typically fused with equal weights [12], [45]. However, through experiments, we find that HSI plays a dominant role in the final prediction. This means that LiDAR data serve as a supplementary source by providing elevation information. As shown in Table V, when using only HSI data, the OA values on Houston2013, MUUFL, and Trento datasets are higher by 78.88%, 43.57%, and 0.39%, respectively, compared with using only LiDAR data. Quantitative evaluation reveals that HSI contributes 65.57% to the overall classification accuracy, significantly outperforming LiDAR (34.43%). Based on this, the MFM module is designed. On the Trento dataset, the MFM module shows the greatest influence, suggesting its effectiveness in fusing LiDAR and HSI data for richer feature representation. Removing SFFE degrades performance across all datasets, demonstrating that parallel spatial-wavelet domain feature extraction outperforms single SFE. In summary, the quantitative analysis confirms that the proposed submodules (CDFM, MFM, and SFFE) significantly enhance MCFNet's ability to process HSI and LiDAR data, leading to superior classification performance.

2) Missing Multimodal Learning: The results of modality elimination are presented in Table V. In order to guarantee fairness, we control the number of parameters in each submodel to be of the same order of magnitude. It is evident that, regardless of which modality is eliminated, there is a certain degree of performance degradation, indicating that the information from multiple modalities can complement each other. Notably, eliminating HSI data seem to cause a more significant decline compared to removing LiDAR data. This suggests that, in the joint classification process, HSI data play a primary role, while LiDAR data serve a supplementary function. Furthermore, to demonstrate the competitiveness of the proposed method in single-modality scenarios, we compare it with two SOTA methods: LRDTN [46] and MamTrans [47]. Ding et al. [46] integrated CNN and Transformer to simultaneously capture local and global relationships in HSI. Shi et al. [47] integrated Mamba and Transformer in a dual-branch structure for single-modal classification. As shown in Table V, even with fewer parameters, our model with HSI outperforms both of these methods on all three datasets. This demonstrates the robustness of our approach and its ability to extend to single-modal methods.

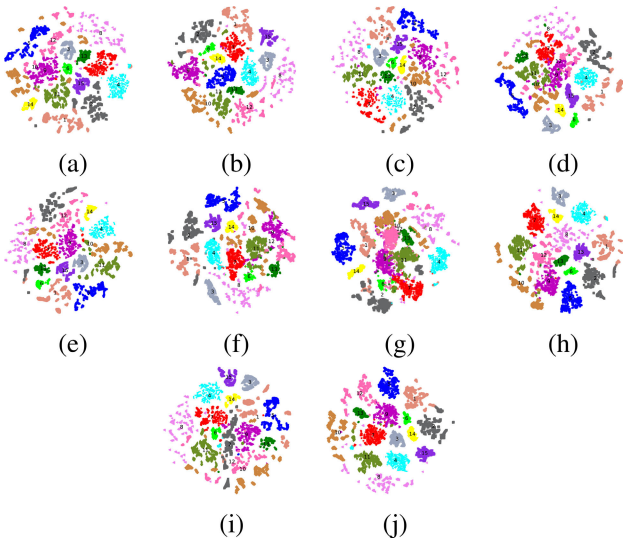


Fig. 8. Feature visualization via t-SNE for testing samples in the Houston2013 dataset. (a) CoupledCNN. (b) GLTNet. (c) FusionHCT. (d) AMSSE. (e) ExViT. (f) HFSNet. (g) HyperMLP. (h) FDNet. (i) M2FNet. (j)MCFNet.

E. Parameter Discussion

1) Visual Analysis: t-SNE employs Gaussian distributions to model the pairwise similarities between data points. By mapping the data to a lower dimensional space, it minimizes the difference between the distribution of points in the lower dimensional space and the distribution of points in the high-dimensional space. Fig. 8 shows the t-SNE visualization

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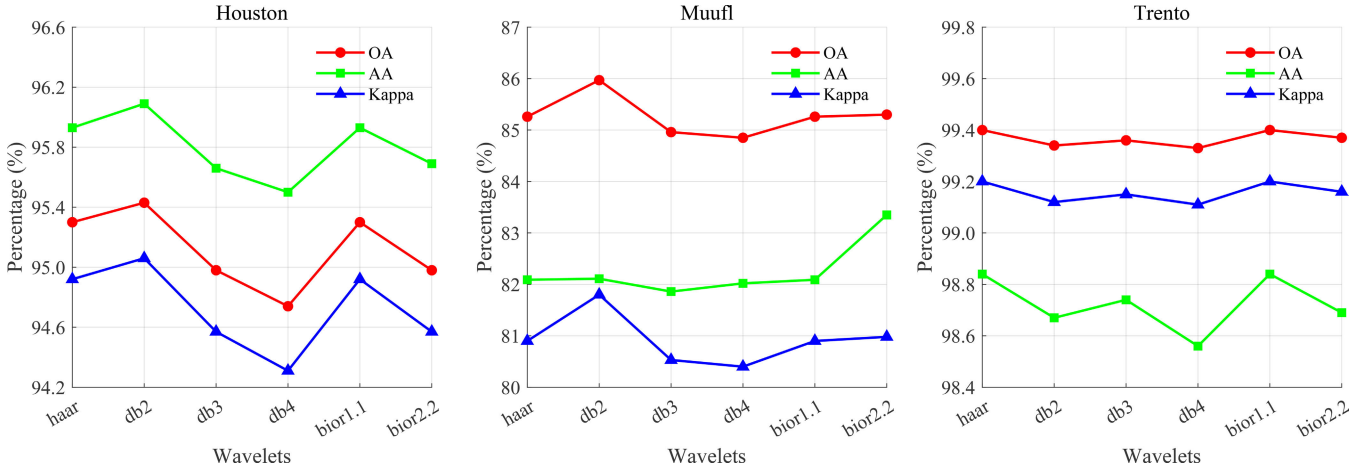


Fig. 9. Quantitative results for different wavelets.

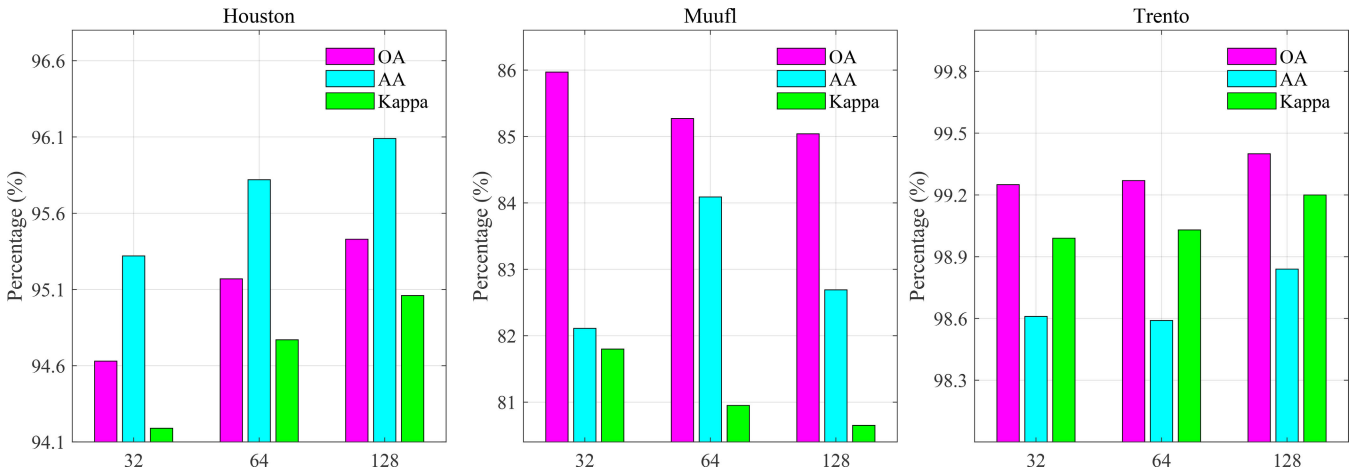


Fig. 10. Quantitative results for a different dimension d_f.

of various methods on the Houston2013 dataset. Notably, this result is consistent with the outcomes presented in Table I. For example, for category 12 (Park Lot 1), many methods perform poorly, and this is also reflected in the visualization. However, our method shows fewer scattered points in this category. Moreover, for other categories, the data points in our method are more concentrated, fully demonstrating the superiority of MCFNet.

2) Different Wavelet Transforms: Different wavelet basis functions provide information about the image at different resolutions. Therefore, we introduce parameter experiments for different wavelet basis functions. As shown in Fig. 9, we use three different wavelet transform: Haar, Daubechies, and Biorthogonal. Haar wavelet demonstrates strong edge preservation in the detail parts, Daubechies has better orthogonality and higher compression efficiency, and Biorthogonal offers better symmetry. As can be seen from Fig. 9, db2 performs the best on both the Houston2013 and Muufl datasets, which we hypothesize is because the images in these two datasets are relatively smooth. On the Trento dataset, Haar and Bior1.1 wavelets perform similarly. Combining this with the results from Table IV, we believe that spatial-domain features are more important for the Trento dataset.

3) Different Dimensions of d_f: We conduct quantization experiments for d_f on three different datasets and observe differences in the results for each dataset, which may be due to the varying scenarios between the datasets. As shown in Fig. 10, it is evident that when d_f is set to 128, the performance is good on both the Houston2013 and Trento datasets, indicating that a larger d_f value allows the network to learn more complex features, which helps capture finer details and high-level information in the data. However, on the Muufl dataset, a smaller d_f achieves good results, balancing both efficiency and accuracy.

V. Conclusion

In this article, a cross-domain model called MCFNet is proposed to effectively fuse different modalities. Specifically, we leverage the advantages of wavelet transforms in both the frequency and time domains to capture texture details of multimodal inputs, while concurrently extracting spatial features from the image. These domain-specific features are then fused for joint classification. We design three key stages: SFFE, cross-domain fusion, and classification. First, SFFE extracts multiscale global frequency-domain and local spatial-domain features. Next, cross-domain fusion enables the fusion

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of spatial- and frequency-domain features, while also achieving the complementary fusion of multimodal features. Finally, in the Classification stage, decision-level fusion is applied to achieve improved classification performance. The effectiveness of our model has been rigorously validated through comprehensive experimental evaluations.

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Qiya Song received the Ph.D. degree in control science and engineering from Hunan University, Changsha, China, in 2023, under the supervision of Prof. Shutao Li.

He is currently a Lecturer with the College of Information Science and Engineering, Hunan Normal University, Changsha. His research interests include multimodal information fusion and computer vision.



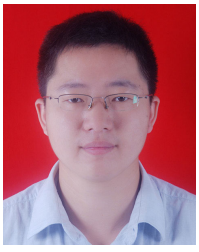
Feng Mo received the B.S. degree in information and computational science from Shanghai Polytechnic University, Shanghai, China, in 2022. She is currently pursuing the M.S. degree in computer science and technology with Hunan Normal University, Changsha, China.

Her research interests include multimodal recommender systems, and hyperspectral image (HSI) and light detection and ranging (LiDAR) data joint classification.



Kexin Ding received the B.S. degree in measurement and control technology and instrument from Liaoning University, Shenyang, Liaoning, China, in 2020. He is currently pursuing the Ph.D. degree in control science and engineering with Hunan University, Changsha, China.

His research interests focus on deep learning and remote sensing data processing.



Lin Xiao received the B.S. degree in electronic information science and technology from Hengyang Normal University, Hengyang, China, in 2009, and the Ph.D. degree in communication and information systems from Sun Yat-sen University, Guangzhou, China, in 2014.

He is currently a Professor with the College of Information Science and Engineering, Hunan Normal University, Changsha, China. His research interests include neural networks, robotics, and intelligent information processing.



Renwei Dian (Member, IEEE) received the B.S. degree from Wuhan University of Science and Technology, Wuhan, China, in 2015, and the Ph.D. degree from Hunan University, Changsha, China, in 2020.

He is currently an Associate Professor with the School of Robotics, Hunan University. His research interests include hyperspectral image super-resolution, image fusion, tensor decomposition, and deep learning. More information can be found on https://github.com/renweidian



Xudong Kang (Senior Member, IEEE) received the B.S. degree from Northeast University, Shenyang, China, in 2007, and the Ph.D. degree from Hunan University, Changsha, China, in 2015.

In 2015, he joined the College of Electrical Engineering, Hunan University. His research interests include hyperspectral feature extraction, image classification, image fusion, and anomaly detection.



Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University. From May 2001 to October 2001, he was a Research Associate with the Department of Computer Science, The Hong Kong University of Science and Technology, Hong Kong. From November 2002 to November 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, London, U.K. From April 2005 to June 2005, he was a Visiting Professor with the Department of Computer Science, The Hong Kong University of Science and Technology. He is currently a Full Professor with the College of Electrical and Information Engineering, Hunan University. He has authored or co-authored over 200 refereed articles. His research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li is a member of the Editorial Board of *Information Fusion* and *Sensing and Imaging*. He received the Second-Grade State Natural Science Award of China in 2019 and two Second-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is currently an Associate Editor of IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT.

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